Raina Bethea Spring 26' Project Proposal

Market Mood
Discovering Stock Market Regimes Using Unsupervised Data Mining

Introduction:

Financial markets do not behave consistently over time. Periods of steady growth and relatively low volatility can be followed by periods of declining prices, increased uncertainty, or rapid market movement. These different conditions are commonly described as market regimes. Identifying these regimes can help analysts better understand how market behavior changes over time and how different financial indicators behave under varying market conditions.

Traditional market analysis often relies on predefined thresholds or previously labeled market periods. However, data mining techniques provide an opportunity to identify patterns directly from historical financial data. In particular, unsupervised learning methods can group observations according to similarities in their characteristics without requiring each observation to be assigned a known market condition beforehand.

This project proposes an unsupervised data mining approach for discovering market regimes using historical U.S. stock market data. Daily market observations will be transformed into features representing characteristics such as returns, rolling returns, volatility, and momentum. Clustering algorithms will then be used to determine whether naturally occurring groups appear in the data and whether those groups represent meaningful market conditions. The resulting clusters will be analyzed and visualized across time to determine how effectively they characterize changes in market behavior.

Motivation:

Financial markets contain large amounts of historical data, but extracting useful patterns from that data can be challenging because market behavior changes over time. A model or assumption that describes a relatively stable market may not describe periods of high volatility or sharp price declines equally well. Identifying these changing conditions is therefore an important part of understanding financial-market behavior.

This project was selected because market regime identification presents a practical application of unsupervised data mining. Unlike supervised classification, where observations already have known labels, market regimes do not necessarily have universally accepted labels. Clustering therefore provides a useful way to investigate whether distinct patterns can be discovered based on the statistical characteristics of market data itself.

Another motivation for this project is to investigate the interpretability of clustering results. Producing clusters alone does not necessarily demonstrate that meaningful market regimes have been identified. The clusters must also exhibit understandable differences in characteristics such as returns, volatility, and momentum. The project will therefore emphasize both the quality of the clustering and the interpretation of the resulting groups.

Problem Statement:

Can unsupervised data mining techniques identify distinct and interpretable market regimes from historical U.S. stock market behavior?

Project Overview:

Historical daily S&P 500 data will serve as the primary representation of U.S. equity-market behavior. The S&P 500 series available through the Federal Reserve Bank of St. Louis's FRED database contains daily closing index values, with FRED currently providing ten years of daily history for the series.

Rather than clustering the raw index level alone, additional features will be engineered from the historical data to describe market behavior. Proposed features include daily percentage return, short-term rolling return, longer-term rolling return, rolling volatility, and momentum. Additional indicators may be incorporated if they improve the analysis without unnecessarily increasing the complexity of the project.

Before clustering, the dataset will be examined for missing observations and other data-quality issues. Features will be standardized so that variables measured on different scales do not disproportionately influence distance-based clustering.

K-Means clustering will serve as the primary clustering algorithm. Several values for the number of clusters (k) will be evaluated rather than selecting the number of regimes arbitrarily. Hierarchical clustering will also be investigated as a comparison method to determine whether different clustering approaches identify similar structures in the data.

The quality of the clustering will be evaluated using metrics such as the silhouette score and within-cluster variation. However, numerical clustering metrics will not be the only evaluation criteria. Each cluster will also be examined according to its average returns, volatility, momentum, and other included features to determine whether it represents a meaningful and interpretable market condition.

The final clusters may correspond to conditions such as:

- relatively stable/low-volatility markets,
- positive-momentum or growth periods,

- sideways or uncertain markets, and
- high-volatility/market-stress periods.

Importantly, those labels will not be assigned beforehand. They will only be applied after examining the characteristics of the clusters produced by the algorithms. That distinction is important because otherwise we'd essentially be deciding the answer before performing the data mining.

The project will also visualize the discovered regimes across the historical S&P 500 timeline. This will make it possible to examine when each regime occurs and whether periods identified as unusual or high-volatility by the clustering approach correspond to visibly different patterns in market behavior.

The primary experiment will compare different values of *k* and clustering approaches. Additional experiments may investigate whether removing or adding selected features changes the resulting regimes. This provides a more meaningful evaluation than simply executing a single clustering algorithm once, which is important because the final-project guidelines require meaningful experiments and a technical contribution beyond simply implementing an existing algorithm.

Proposed Schedule:

Now Initial Dev	- Data collection and preprocessing - Exploratory Analysis - Feature Engineering
Analysis & Testing	- Initial clustering experiments and analysis
Early November Interim Presentation	- Construct a presentation based on notes and findings. - Present progress of the project and receive feedback.
Expansion & Revision	- Further evaluation, interpretation, and visualization of market regimes
Nov 20 Final Presentation	- Final analysis, report, code/documentation, and project demonstration